

CSS Selectors

In a CSS file, selectors are used to represent parts of your HTML so that you can target specific elements and apply style rules to them.

You can target elements by their **tag names**.

Element Selectors

```
p {  
  line-height: 1.5;  
  padding: 10px;  
}  
img {  
  width: 150px;  
  display: block;  
  margin: 0 auto;  
}
```

You can also use **class** and **ID selectors** to target specific elements. Only elements with a matching id or class will be styled.

For example, the id of content and the class highlight are being targeted in the example.

Class & ID Selectors

```
#content {  
  max-width: 600px;  
  margin: 100px auto;  
}  
.highlight {  
  color: #006EB6;  
  font-size: .75em;  
}
```

Descendant Selectors

Descendant selectors target elements **within** other elements.

For instance:

- Paragraphs inside a <div>
- Elements with the class highlight inside a <section>

Only those nested elements will be affected.

Descendant Selectors

```
div p {  
  line-height: 1.5;  
  padding: 10px;  
}  
section .highlight {  
  color: #006EB6;  
  font-size: .75em;  
}
```

Group Selectors

With **group selectors**, you can target multiple elements at once by separating them with commas.

For example, all <h1>, <h2>, <p>, and elements can be grouped and styled together.

Group Selectors

```
h1, h2 {  
  text-align: center;  
  letter-spacing: 4px;  
}  
p, li {  
  line-height: 1.5;  
  padding: 10px;  
}
```

Pseudo-Class Selectors

Pseudo-class selectors let you target elements in specific states or positions.

For example:

- `a:hover` targets only anchor tags that are being hovered over.
- `li:nth-of-type(3)` targets the **third** list item of its type.

Pseudo-class Selectors

```
a:hover {  
  background-color:  
  beige;  
  color: black;  
}  
li:nth-of-type(3) {  
  background-color:  
  rgba(200, 210, 220, .5)  
}
```

Descendant Combinators: A Closer Look

Let's take a closer look at **descendant combinators**.

div p

```
<body>
  <div>
    <h1>This is the main title</h1>
    <p>This is the main paragraph after the h1.</p>
    <section>
      <p>The paragraph inside the section.</p>
    </section>
    <p>This is the follow-up paragraph</p>
  </div>
</body>
```

```
/* parent descendant */
div p {
  background-color: lightblue;
}
```

This is the main title

This is the main paragraph after the h1.

The paragraph inside the section.

This is the follow-up paragraph

This targets all `<p>` elements **inside** a `<div>` (not just direct children—any level of nesting).

Let's look at some you many have not seen. Here we see the selector:

div > p

```
<body>
  <div>
    <h1>This is the main title</h1>
    <p>This is the main paragraph after the h1.</p>
    <section>
      <p>The paragraph inside the section.</p>
    </section>
    <p>This is the follow-up paragraph</p>
  </div>
</body>
```

```
/* parent direct children */
div > p {
  background-color: lightblue;
}
```

This is the main title

This is the main paragraph after the h1.

The paragraph inside the section.

This is the follow-up paragraph

This targets only `<p>` elements that are **direct children** of a `<div>`. Notice that a `<p>` inside a nested `<section>` wouldn't be selected because it's a **grandchild**, not a direct child.

Here we see the selector:

h1 + p

```
<body>
  <div>
    <h1>This is the main title</h1>
    <p>This is the main paragraph after the h1.</p>
    <section>
      <p>The paragraph inside the section.</p>
    </section>
    <p>This is the follow-up paragraph</p>
  </div>
</body>
```

```
/* immediate next sibling */
h1 + p {
  background-color: lightblue;
}
```

This is the main title
This is the main paragraph after the h1.
The paragraph inside the section.
This is the follow-up paragraph

This targets the **immediate sibling**—the first `<p>` that comes **right after** an `<h1>`.

Here we see the selector:

h1 ~ p

```
<body>
  <div>
    <h1>This is the main title</h1>
    <p>This is the main paragraph after the h1.</p>
    <section>
      <p>The paragraph inside the section.</p>
    </section>
    <p>This is the follow-up paragraph</p>
  </div>
</body>
```

```
/* all sibling that come after */
h1 ~ p {
  background-color: lightblue;
}
```

This is the main title
This is the main paragraph after the h1.
The paragraph inside the section.
This is the follow-up paragraph

This targets **all sibling <p> elements** that come **after** an `<h1>`.

More Pseudo-Class Selectors

Here are some additional pseudo-class selectors:

- `p:first-child` — targets a `<p>` only if it is the **first child** of its parent.
- `p:last-child` — targets a `<p>` only if it is the **last child**.
- `li:nth-child(odd)` — targets all **odd-numbered** `<li>` elements (1st, 3rd, 5th, etc).
- `input:focus` — targets an `<input>` element **only when it is focused**.

There are many more!

For a full reference, check out:

 https://www.w3schools.com/cssref/css_ref_pseudo_classes.php

CSS Attribute Selectors

Attribute selectors let you target elements based on their attributes in the HTML.

Examples:

- `a[target="_blank"]` — targets all `<a>` tags with a target attribute set to `_blank`.
- `input[type="text"]` — targets only `<input>` elements where the type attribute is `"text"`.